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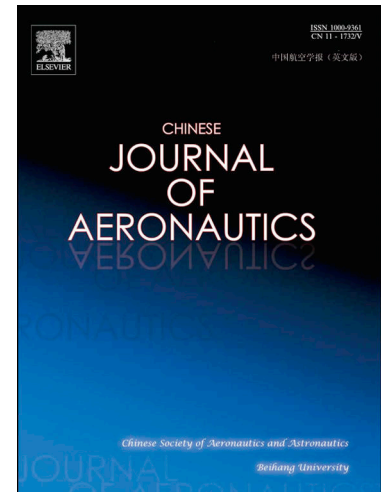
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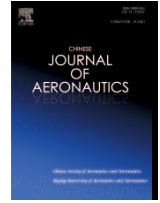
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Uncertainty-aware model predictive control with learning-based residual dynamics compensation for agile quadrotor trajectory tracking

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Abstract

Achieving high-precision trajectory tracking for quadrotors during high-speed and agile maneuvers remains a significant challenge, primarily due to uncertainty induced by model inaccuracies and external disturbances. To address this issue, we propose an Uncertainty-aware Learning-based Model Predictive Control (UL-MPC) method. Specifically, we learn residual dynamics using a probabilistic neural network that outputs both the residual mean and covariance. The learned mean compensates the nominal model, while the predicted uncertainty is incorporated into chance-constrained MPC via covariance propagation and adaptive constraint tightening, leading to improved tracking accuracy while enhancing robust constraint satisfaction. Furthermore, we provide a Lyapunov-based analysis to establish local mean-square boundedness of the linearized closed-loop error dynamics. High-fidelity Gazebo simulations with stochastic wind and aggressive trajectories demonstrate consistent improvements over nominal MPC and state-of-the-art learning-based MPC baselines, achieving approximately a 32% reduction in tracking errors and a 44% reduction in safety-boundary violation rate relative to the strongest baseline.

Keywords: Model predictive control; Learning-based control; Trajectory tracking; Unmanned aerial vehicles; Drones

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1. Introduction

As the application of quadrotors expands to high-speed and agile flight scenarios, the need for accurate trajectory tracking becomes increasingly important. However, trajectory tracking control in such conditions remains hindered by multiple sources of uncertainty, including modeling inaccuracies in the dynamics^{1,2} and external wind disturbances³. These effects can lead to significant tracking deviations and complicate controller design. Consequently, maintaining precise trajectory tracking under uncertainty conditions remains a critical research challenge.

Model Predictive Control (MPC) has gained prominence in aerial robotics due to its ability to optimize control while handling multiple constraints and predicting future behavior. However, the effectiveness of MPC fundamentally depends on the accuracy of the underlying dynamics model⁴. Accurately modeling quadrotor dynamics becomes increasingly difficult under high-speed motion, rapid acceleration, or wind disturbances. Specifically, the uncertainty arises from aerodynamic effects such as airflow-induced lift and drag and propeller-downwash turbulence, while external wind disturbances further complicate the modeling⁵.

To improve system identification accuracy in the presence of complex dynamics, learning-based MPC⁶ has been widely studied. Previous work typically focuses on augmenting the nominal dynamic model with a data-driven model, such as Gaussian processes², policy search⁷ or neural networks⁸⁻¹⁰, to improve prediction accuracy. In real flight, high-speed tracking and wind rejection are safety-critical because unmodeled aerodynamics and gust transients may render nominally feasible plans unsafe, leading to constraint violations and repeated corrective actions. This practical need has motivated learning-based residual compensation and uncertainty-aware constraint handling in MPC-based quadrotor tracking studies^{10,11}. Although these methods have shown effectiveness in agile flight, most offer limited treatment of uncertainty. In particular, predictive uncertainty is often ignored or approximated through conservative bounds, which restricts robustness under uncertainty. The high-capacity learning-based MPC with effective uncertainty handling remains challenging for agile flight applications.

In this paper, we propose an Uncertainty-aware Learning-based Model Predictive Control (UL-MPC) method for quadrotor trajectory tracking in agile flights. Firstly, the method leverages a dual-head neural network to jointly compensate residual dynamics and estimate the uncertainty arising from unmodeled aerodynamic effects. Moreover, by incorporating the learned uncertainty into a chance-constrained MPC formulation, UL-MPC enables adaptive constraint tightening to enhance robustness against disturbances. The proposed method is validated in high-fidelity quadrotor simulations under wind disturbances and aggressive maneuvers. Compared to nominal MPC and learning-based MPC baselines^{2,8}, our approach demonstrates superior tracking accuracy and enhanced robustness.

The main contributions of this work are as follows:

- (1) A probabilistic residual dynamics compensation model is proposed, where a dual-head neural network is employed to jointly estimate the mean and uncertainty of unmodeled aerodynamic effects in agile quadrotor flights.
- (2) An uncertainty-aware learning-based model predictive control method for quadrotor trajectory tracking is developed, which incorporates the predicted uncertainty into a chance-constrained optimization formulation, enabling risk-sensitive constraint tightening under uncertainty.
- (3) A Lyapunov-based analysis is provided to establish local mean-square boundedness of the linearized closed-loop tracking error under bounded residual uncertainty. Furthermore, high-fidelity simulations under aggressive flight and wind disturbances confirm superior tracking accuracy and robustness compared to standard MPC and learning-based baselines.

2. Related work

Ensuring robust and accurate control of agile quadrotor flight remains a fundamental challenge, primarily due to complex uncertainties that are further amplified in disturbed environments. This section provides an overview of recent developments in uncertainty-aware control and learning-based model predictive control.

2.1. Uncertainty-aware control for agile flight

The growing impact of unmodeled dynamics and external disturbances in agile flight underscores the need for uncertainty-aware control strategies.

Classical robust control methods, including H_∞ ¹², Proportional-Integral-Derivative (PID)¹³, and disturbance observer-based control¹⁴, have been employed to attenuate the effects of bounded disturbances and modeling errors. Adaptive control approaches, such as L1 adaptive control¹⁵, also offer online model adjustment capabilities for handling time-varying dynamics and wind-induced deviations. However, these methods demonstrate critical limitations during high-speed agile maneuvers.

To address this, recent research has turned to use data-driven technique to deal with uncertainty estimation problem. For example, Gaussian Processes (GPs) provide a Bayesian nonparametric framework for modeling both mean dynamics and predictive variance^{16,17}, enabling chance-constrained MPC formulations for safe control under wind disturbances. Nevertheless, the cubic scaling of GP inference with respect to dataset size and dimensionality introduces prohibitive computational costs, particularly under the resource constraints of onboard embedded platforms. Motivated by these challenges, neural network-based approximations have gained momentum owing to their scalability and expressive capacity. Ensemble methods¹⁸ and dropout-based Bayesian approximations¹⁹ allow for approximate uncertainty, and have been integrated into control pipelines via stochastic MPC²⁰ or probabilistic residual learning²¹. However, many of these methods involve repeated sampling or multi-model inference, introducing latency that limits their applicability in high-speed control loops.

More recent studies have explored hybrid schemes that couple learned uncertainty representations with structured optimization-based control. Belief-space formulations²² incorporate state-estimation uncertainty by optimizing over a belief representation, but this typically increases problem dimensionality and thus the online computational burden. In parallel, physics-informed learning-augmented MPC approaches, such as RAMP-Net²³, embed physical priors into the learning module to improve robustness while preserving real-time solvability, offering a different accuracy-latency trade-off. Nie et al.²⁴ proposed a model-free adaptive optimal control framework for rocket-powered vehicles, where a dynamic neural network approximates uncertain dynamics within an adaptive dynamic programming scheme, leading to strong adaptability to aerodynamic disturbances. Collectively, these developments underscore the importance of uncertainty-aware control in ensuring robust, accurate, and safe trajectory tracking under wind disturbances, which is essential for agile and reliable quadrotor operation in dynamic aerial environments.

2.2. Learning-based model predictive control

The emergence of deep learning has sparked growing interest in harnessing the expressive capacity of neural networks for system modeling and control. A prominent strategy in this paradigm is residual dynamics compensation²⁵, where neural networks or Gaussian Processes are trained to approximate the residual between nominal dynamics models and the true system^{17,26}. By embedding these residual models into the MPC prediction loop, model mismatch can be effectively mitigated, thereby enhancing tracking accuracy and robustness^{27,28}. Notably, Bauersfeld et al.¹ introduced NeuroBEM, a hybrid framework that augments blade-element momentum theory with learned residuals to model aerodynamic thrust, torques, and parasitic effects in agile flight. Kulathunga et al.²¹ proposed a GP-based MPC scheme capable of robust trajectory tracking under distributional shifts in dynamics of quadrotors.

Recent developments in learning-based MPC highlight the importance of uncertainty estimation to improve robustness and satisfy constraints. Since learned models can have errors caused by limited data or incorrect assumptions, estimating predictive uncertainty becomes crucial for safe and reliable control. Kulathunga et al.²⁵ incorporated uncertainty into GP-based residual models, while Depeweg et al.²⁹ proposed a neural network method that tracks how prediction variance evolves over the MPC horizon. These approaches are particularly effective in chance-constrained MPC, where accounting for uncertainty is key to ensuring constraint satisfaction. Beyond residual modeling, some recent work has explored full system identification using flexible learning models. Instead of relying on physics-based equations, these methods learn the entire dynamics from data^{30,31}. While this allows for greater modeling flexibility, it often requires large amounts of data and careful training to maintain closed-loop stability.

Inspired by promising advances in previous work^{2,8}, this work introduces a probabilistic residual model that jointly estimates the predictive mean and variance of unmodeled dynamics. When embedded in a chance-constrained MPC scheme, the model enables scalable, risk-sensitive control while maintaining real-time feasibility. By integrating residual dynamics compensation and uncertainty estimation within a unified predictive control pipeline, the proposed method improves agile quadrotor performance under aerodynamic disturbances.

3. Method

The proposed Uncertainty-aware Learning-based MPC (UL-MPC) method integrates first-principles dynamics with data-driven uncertainty modeling in a receding-horizon optimal control formulation, as illustrated in Fig. 1. Specifically, the framework combines chance-constrained MPC with a residual dynamics model to process quadrotor states and reference trajectories. A learning-based Multilayer Perceptron (MLP) module with a dual-head architecture jointly predicts the mean (used to compensate the residual dynamics) and the variance (used for uncertainty quantification). These predictions are incorporated into the chance-constrained MPC, enabling adaptive constraint tightening and improved robustness against wind disturbances.

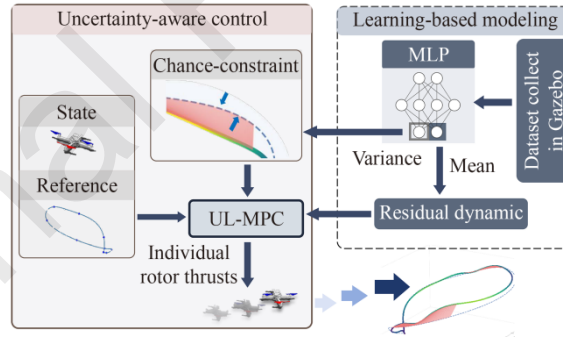


Fig. 1 Architecture of UL-MPC.

3.1. System modeling and problem statement

We model the quadrotor as a rigid body evolving in three-dimensional space with six degrees of freedom. The system dynamics are governed by the Newton-Euler equations, with the state vector defined as:

$$\mathbf{x} = [\mathbf{p}_{WB}^T, \mathbf{v}_{WB}^T, \mathbf{q}_{WB}^T, \mathbf{\omega}_B^T]^T \in \mathbb{R}^{13} \quad (1)$$

where $\mathbf{p}_{WB} \in \mathbb{R}^3$ and $\mathbf{v}_{WB} \in \mathbb{R}^3$ denote the position and velocity in the world frame, $\mathbf{q}_{WB} \in \mathbb{R}^3$ represents the unit quaternion for orientation, and $\mathbf{\omega}_B \in \mathbb{R}^3$ is the angular velocity expressed in the body frame.

The control input $\mathbf{u} \in \mathbb{R}^4$ consists of the individual rotor thrusts $\{T_0, T_1, T_2, T_3\}$. The nominal continuous-time dynamics are given by

$$\dot{\mathbf{x}} = f_{\text{nom}}(\mathbf{x}, \mathbf{u}) = \begin{bmatrix} \mathbf{v}_{WB} \\ \frac{1}{2} \mathbf{q}_{WB} \cdot \begin{bmatrix} 0 \\ \boldsymbol{\omega}_B \end{bmatrix} \\ \frac{1}{m} \mathbf{q}_{WB} \mathbf{e} T_B + \mathbf{g}_W \\ \mathbf{J}^{-1}(\boldsymbol{\tau}_B - \boldsymbol{\omega}_B \times \mathbf{J} \boldsymbol{\omega}_B) \end{bmatrix} \quad (2)$$

where m is the quadrotor mass, $\mathbf{J} = \text{diag}(J_x, J_y, J_z)$ is the inertia matrix, $\mathbf{g}_W = [0, 0, -9.81]^T$ denotes gravity in the world frame, $\mathbf{e}$ represents the transformation of thrust from the body to the world frame via quaternion, and $\cdot$ denotes quaternion multiplication.

The total thrust and torque are computed from the individual rotor inputs as:

$$\mathbf{T}_B = \begin{bmatrix} 0 \\ 0 \\ \sum_{i=0}^3 T_i \end{bmatrix}, \quad \boldsymbol{\tau}_B = \begin{bmatrix} d_y(-T_0 - T_1 + T_2 + T_3) \\ d_x(-T_0 + T_1 + T_2 - T_3) \\ c_\tau(-T_0 + T_1 - T_2 + T_3) \end{bmatrix} \quad (3)$$

where d_x, d_y are rotor arm lengths and c_τ denotes the torque coefficient from drag.

For controller design, the continuous-time quadrotor dynamics are discretized using a fourth-order Runge-Kutta (RK4) integrator with step size Δt , yielding the nominal discrete-time model

$$\mathbf{x}_{k+1} = f_{\text{RK4}}(\mathbf{x}_k, \mathbf{u}_k, \Delta t) \quad (4)$$

where $\mathbf{x}_k \in \mathbb{R}^{13}$ denotes the system state at step k , and $\mathbf{u}_k \in \mathbb{R}^4$ is the control input.

Let $\mathbf{x}_k^{\text{ref}}$ and $\mathbf{u}_k^{\text{ref}}$ denote the reference state and input trajectories over a finite horizon of length N . The standard MPC problem seeks an input sequence $\mathbf{u}_{0:N-1}$ that minimizes the finite-horizon tracking cost:

$$\begin{aligned} & \min_{\mathbf{u}_{0:N-1}} \sum_{k=0}^{N-1} l(\mathbf{x}_k, \mathbf{u}_k) \\ & \text{s.t.} \quad \begin{cases} \mathbf{x}_{k+1} = f_{\text{RK4}}(\mathbf{x}_k, \mathbf{u}_k, \Delta t) \\ \mathbf{x}_0 = \mathbf{x}_{\text{init}} \\ \mathbf{x}_k \in \mathbf{X} \\ \mathbf{u}_k \in \mathbf{U} \end{cases} \end{aligned} \quad (5)$$

with the quadratic stage cost defined as

$$l(\mathbf{x}_k, \mathbf{u}_k) = \|\mathbf{x}_k - \mathbf{x}_k^{\text{ref}}\|_Q^2 + \|\mathbf{u}_k - \mathbf{u}_k^{\text{ref}}\|_R^2 \quad (6)$$

where $\mathbf{X}$ and $\mathbf{U}$ denote the admissible state and input sets, respectively, and $\mathbf{x}_{\text{init}}$ is the initial condition.

However, the nominal model does not account for several critical real-world effects that become increasingly prominent during high-speed and aggressive flight. These effects induce structured modeling errors that depend on the system state and control input, and cannot be adequately captured by fixed uncertainty bounds or simple additive noise assumptions. To explicitly address such discrepancies, we introduce a learning-based residual dynamics model in the following subsection.

3.2. Learning-based residual dynamics modeling

To account for discrepancies between the nominal model and the true quadrotor dynamics, we introduce a learning-based residual modeling approach that explicitly characterizes the one-step discrete-time modeling error. Starting from the nominal discrete-time dynamics obtained via numerical integration, the true system evolution can be expressed as:

$$\mathbf{x}_{k+1} = f_{\text{nom}}(\mathbf{x}_k, \mathbf{u}_k) + f_{\Delta}(\mathbf{x}_k, \mathbf{u}_k) \quad (7)$$

where f_{nom} denotes the nominal dynamics, and f_{Δ} represents the residual dynamics capturing unmodeled effects.

3.2.1 Residual definition and dataset

The residual term $f_{\Delta}(\mathbf{x}, \mathbf{u})$ aggregates structured modeling errors arising from multiple sources, including neglected aerodynamic drag, actuator dynamics mismatches, and external disturbance such as wind. These effects are generally state- and input-dependent and become more pronounced during aggressive maneuvers and high-speed flight. Rather than explicitly modeling each source, we treat their combined influence as a structured but uncertain residual learned directly from data. Concretely, given a transition tuple $(\mathbf{x}_k, \mathbf{u}_k, \mathbf{x}_{k+1}^{\text{true}})$, the empirical one-step residual target is defined as

$$\Delta \mathbf{x}_k @ \mathbf{x}_{k+1}^{\text{true}} - f_{\text{nom}}(\mathbf{x}_k, \mathbf{u}_k) \quad (8)$$

and the training dataset is formed as $\mathcal{D} = \{(\mathbf{x}_k, \mathbf{u}_k, \Delta \mathbf{x}_k)\}$.

3.2.2. Probabilistic residual model

We model the residual dynamics using a stochastic function approximator $f_{\Delta} \in \mathcal{F}_{\theta}$ instantiated by a feedforward multilayer perceptron (MLP) with parameters θ . The network outputs a predictive mean $\mu_{\Delta, \theta}(\mathbf{x}, \mathbf{u}) \in \mathbb{R}^n$ and a state-dependent diagonal variance $\sigma_{\Delta, \theta}^2(\mathbf{x}, \mathbf{u}) \in \mathbb{R}_+^n$, yielding the heteroscedastic Gaussian representation

$$f_{\Delta}(\mathbf{x}, \mathbf{u}) \sim \mathcal{N}(\mu_{\Delta, \theta}(\mathbf{x}, \mathbf{u}), \Sigma_{\Delta, \theta}(\mathbf{x}, \mathbf{u})), \quad \Sigma_{\Delta, \theta}(\mathbf{x}, \mathbf{u}) @ \text{diag}(\sigma_{\Delta, \theta}^2(\mathbf{x}, \mathbf{u})) \quad (9)$$

This Gaussian assumption is adopted as a tractable moment-matching surrogate that summarizes residual

uncertainty by its first- and second-order statistics, enabling efficient integration into the downstream uncertainty-aware MPC. In the proposed controller, the mean $\mu_{\Delta,\theta}$ is used to compensate the nominal dynamics for improved prediction accuracy, while the variance $\sigma_{\Delta,\theta}^2$ quantifies residual uncertainty and will be exploited for adaptive constraint tightening.

3.2.3. Training objective and implementation.

The network parameters θ are learned by minimizing the Negative Log-Likelihood (NLL) of observed residuals under the predicted Gaussian distribution:

$$\mathcal{L}_{\text{NLL}} = \frac{1}{|\mathcal{D}|} \sum_{k=1}^{|\mathcal{D}|} \sum_{i=1}^n \left[\frac{1}{2} \ln(2\pi\sigma_{\Delta,k,i}^2) + \frac{(\Delta\mathbf{x}_{k,i} - \mu_{\Delta,k,i})^2}{2\sigma_{\Delta,k,i}^2} \right] \quad (10)$$

where $\Delta\mathbf{x}_{k,i}$ denotes the observed one-step residual in the i -th state dimension, and $(\mu_{\Delta,k,i}, \sigma_{\Delta,k,i}^2)$ are the corresponding network predictions evaluated at $(\mathbf{x}_k, \mathbf{u}_k)$.

Training data are collected from high-fidelity Gazebo simulations, where the residual captures the combined effects of unmodeled physical dynamics and simulation modeling discrepancies. To promote generalization across operating conditions, the dataset is collected from diverse rollouts with randomized waypoint trajectories and injected wind disturbance of varying magnitudes and directions. Standard regularization techniques are applied during training, including dropout and ℓ_2 weight decay to mitigate overfitting and prevent memorization of specific trajectories.

Importantly, UL-MPC does not assume perfect generalization of the learned residual model to arbitrarily Out-Of-Distribution (OOD) disturbances. Under extreme or unseen conditions, the residual uncertainty is expected to increase, which in turn yields a more conservative behavior through uncertainty-aware constraint tightening; however, high-precision tracking is not guaranteed in such OOD regimes.

For real-time feasibility in MPC, we adopt the diagonal covariance structure in Eq.(9), which reduces the computational burden in uncertainty propagation and constraint tightening; the framework can be extended to structured covariance parameterizations at increased computational cost.

3.3. Uncertainty-aware learning-based MPC

To ensure reliable trajectory tracking in the presence of unmodeled dynamics and stochastic disturbances, we formulate an UL-MPC scheme that explicitly incorporates the learned residual uncertainty into constraint handling. The key objective is to mitigate the risk of state constraint violations induced by prediction errors, while retaining the computational structure of a deterministic MPC.

Unlike standard MPC formulations that rely on a fixed feasible set and implicitly assume perfect model knowledge, UL-MPC adapts its effective state constraints online according to the predicted residual uncertainty. This adaptive tightening enables a principled trade-off between tracking performance and safety, particularly in regions of the state space where modeling uncertainty is elevated.

3.3.1. Augmented dynamics and uncertainty-aware constraints

We employ the augmented discrete-time dynamics in Eq.(7), where the nominal dynamics are compensated using the predictive mean of the learned residual model. Denoting the predicted state mean by $\bar{\mathbf{x}}_k$ over the MPC horizon, we use

$$\bar{\mathbf{x}}_{k+1} = f_{\text{nom}}(\bar{\mathbf{x}}_k, \mathbf{u}_k) + \boldsymbol{\mu}_{\Delta, \theta}(\bar{\mathbf{x}}_k, \mathbf{u}_k) \quad (11)$$

where $\boldsymbol{\mu}_{\Delta, \theta}(\cdot)$ is defined in Section 3.2. The residual uncertainty is modeled as an additive stochastic term whose covariance is given by Eq.(9).

To obtain uncertainty-aware safety margins along the horizon, we propagate the state prediction covariance $\boldsymbol{\Sigma}_k$ using a first-order approximation described in Section 3.4.

In the presence of uncertainty, enforcing hard state constraints of the form $\mathbf{x}_k \in \mathbf{X}$ may lead to systematic constraint violations. To explicitly control the risk of unsafe behavior, we impose chance constraints on each constrained state component:

$$\mathbb{P}(x_{k,i} \leq x_i^{\max}) \geq 1 - \delta, \quad \delta \in (0, 1) \quad (12)$$

Under the Gaussian moment approximation in Section 3.2, the i -th marginal of the predicted state is characterized by mean $\bar{x}_{k,i}$ and variance $[\boldsymbol{\Sigma}_k]_{ii}$. Define the corresponding marginal standard deviation as

$$\sigma_{x,k,i} @ \sqrt{[\boldsymbol{\Sigma}_k]_{ii}} \quad (13)$$

A conservative deterministic sufficient condition for Eq.(12) is then

$$\bar{x}_{k,i} + \beta_{\delta} \sigma_{x,k,i} \leq x_i^{\max} \quad (14)$$

where β_{δ} is the confidence parameter associated with the risk level δ . We choose β_{δ} as the $(1 - \delta)$ -quantile of a standard normal random variable Z , i.e., $\mathbb{P}(Z \leq \beta_{\delta}) = 1 - \delta$. Here, we use $\delta = 0.05$ as a default choice, resulting in $\beta_{\delta} \approx 1.645$.

The inequality Eq.(14) can be interpreted as an uncertainty-dependent safety margin that shrinks the admissible set in regions of high modeling uncertainty. In practice, the chance-constrained adjustment is computed in a component-wise manner. At each prediction step k , we first obtain the covariance $\boldsymbol{\Sigma}_k$ from the recursion in Section 3.4, then compute $\sigma_{x,k,i} = \sqrt{[\boldsymbol{\Sigma}_k]_{ii}}$, and finally tighten the bound by replacing $x_{k,i} \leq x_i^{\max}$ with $\bar{x}_{k,i} \leq x_i^{\max} - \beta_{\delta} \sigma_{x,k,i}$. The tightened bounds are directly used as constraints in the UL-MPC optimization Eq.(15).

3.3.2. Chance-constrained MPC formulation

The UL-MPC problem is formulated as the following finite-horizon optimization:

$$\begin{aligned} \min_{\mathbf{u}_{0:N-1}} & \sum_{k=0}^{N-1} l(\bar{\mathbf{x}}_k, \mathbf{u}_k) \\ \text{s.t.} & \begin{cases} \bar{\mathbf{x}}_{k+1} = f_{\text{nom}}(\bar{\mathbf{x}}_k, \mathbf{u}_k) + \boldsymbol{\mu}_{\Delta, \theta}(\bar{\mathbf{x}}_k, \mathbf{u}_k) \\ \boldsymbol{\Sigma}_{k+1} = \mathbf{G}(\boldsymbol{\Sigma}_k, \bar{\mathbf{x}}_k, \mathbf{u}_k, \boldsymbol{\Sigma}_{\Delta, k}) \\ \bar{x}_{k,i} + \beta_{\delta} \sigma_{x,k,i} \leq x_i^{\max}, \quad \forall i, k \end{cases} \end{aligned} \quad (15)$$

where $\mathbf{X}$ and $\mathbf{U}$ denote the deterministic admissible state and input sets, respectively, and $\mathbf{G}(\cdot)$ is the covariance propagation mapping induced by Eq.(20).

The uncertainty-affected state bounds are enforced via the tightening in Eq.(14). The residual variance does not modify the mean dynamics Eq.(11); it influences the solution only through the tightened constraints, making the controller more conservative as the predicted uncertainty increases. When the predicted uncertainty becomes large, the tightening margins can significantly shrink the feasible set, and feasibility may not be preserved under arbitrarily large out-of-distribution uncertainties.

3.3.3. UL-MPC algorithm

The resulting UL-MPC follows a standard receding-horizon structure, combining offline residual learning with online chance-constrained MPC, as outlined in Algorithm 1. In the offline phase, residuals obtained from nominal MPC rollouts are used to train the probabilistic residual model. During online execution, the learned residual mean and variance are evaluated within the MPC prediction model and constraint tightening along the horizon, and the first control input of the optimal sequence is applied in a receding-horizon manner.

Algorithm 1. UL-MPC

Input: Initial state $\mathbf{x}_0$, reference trajectory $\{\mathbf{x}_k^{\text{ref}}\}_{k=0}^N$

Output: Applied control inputs $\{\mathbf{u}_t^*\}$

// Offline: probabilistic residual model training

1. Initialize dataset $\mathbf{D} \leftarrow \emptyset$
2. **for** each rollout under nominal MPC **do**
3. Record $\{\mathbf{x}_k, \mathbf{u}_k, \mathbf{x}_{k+1}^{\text{true}}\}$
4. Compute residual $\Delta \mathbf{x}_k = \mathbf{x}_{k+1}^{\text{true}} - f_{\text{nom}}(\mathbf{x}_k, \mathbf{u}_k)$
5. Append $(\mathbf{x}_k, \mathbf{u}_k, \Delta \mathbf{x}_k)$ to $\mathbf{D}$
6. Train the dual-head residual network to obtain $(\boldsymbol{\mu}_{\Delta, \theta}, \boldsymbol{\Sigma}_{\Delta})$ via the NLL loss Eq.(10)

// Online: receding-horizon UL-MPC

7. Initialize $\{\mathbf{u}_{\tau|0}\}_{\tau=0}^{N-1}$ (warm start)
 8. **for** each replanning update $k = 0, 1, \dots$ **do**
 9. Shift the previous optimal solution to warm-start $\{\mathbf{u}_{k+\tau|k}\}$
 10. Predict the mean trajectory using Eq.(11)
 11. Evaluate $(\boldsymbol{\mu}_{\Delta,\theta}, \boldsymbol{\Sigma}_{\Delta,k})$ along the predicted trajectory
 12. Propagate $\boldsymbol{\Sigma}_k$ along the horizon using Eq.(20)
 13. Compute tightened constraints using Eq.(12)-(14)
 14. Solve Eq.(15) with SQP-RTI to obtain $\{\mathbf{u}_{k+\tau|k}^*\}_{\tau=0}^{N-1}$
 15. Apply $\mathbf{u}_{k|k}^*$ and observe $\mathbf{x}_{k+1}$
-

3.4. Implementation

For computational tractability, we employ a first-order approximation to propagate uncertainty locally around the predicted trajectory. Real-time feasibility is achieved by embedding the resulting recursion into a CasADi-based optimization framework³². In the proposed UL-MPC, the predicted mean dynamics are given by the augmented model

$$\bar{\mathbf{x}}_{k+1} = f_{\text{nom}}(\bar{\mathbf{x}}_k, \mathbf{u}_k) + \boldsymbol{\mu}_{\Delta,\theta}(\bar{\mathbf{x}}_k, \mathbf{u}_k) \quad (16)$$

and the residual uncertainty is modeled as an additive disturbance with covariance

$$\boldsymbol{\Sigma}_{\Delta,k} @ \text{diag}(\sigma_{\Delta,\theta}^2(\bar{\mathbf{x}}_k, \mathbf{u}_k)) \quad (17)$$

Let $f(\mathbf{x}, \mathbf{u}) @ f_{\text{nom}}(\mathbf{x}, \mathbf{u}) + \boldsymbol{\mu}_{\Delta,\theta}(\mathbf{x}, \mathbf{u})$ denote the augmented mean dynamics. Linearizing f around the predicted mean trajectory $(\bar{\mathbf{x}}_k, \mathbf{u}_k)$ yields

$$\mathbf{x}_{k+1} \approx f(\bar{\mathbf{x}}_k, \mathbf{u}_k) + \mathbf{A}_k(\mathbf{x}_k - \bar{\mathbf{x}}_k) \quad (18)$$

where

$$\mathbf{A}_k @ \frac{\partial f}{\partial \mathbf{x}} \bigg|_{(\bar{\mathbf{x}}_k, \mathbf{u}_k)} = \frac{\partial f_{\text{nom}}}{\partial \mathbf{x}} \bigg|_{(\bar{\mathbf{x}}_k, \mathbf{u}_k)} + \frac{\partial \boldsymbol{\mu}_{\Delta,\theta}}{\partial \mathbf{x}} \bigg|_{(\bar{\mathbf{x}}_k, \mathbf{u}_k)} \quad (19)$$

In practice, $\mathbf{A}_k$ is obtained efficiently via CasADi automatic differentiation. Under this approximation and assuming the residual uncertainty enters additively with covariance Eq.(17), the predicted state covariance evolves as

$$\boldsymbol{\Sigma}_{k+1} = \mathbf{A}_k \boldsymbol{\Sigma}_k \mathbf{A}_k^T + \boldsymbol{\Sigma}_{\Delta,k}, \quad \boldsymbol{\Sigma}_0 = 0 \quad (20)$$

The recursion Eq.(20) is evaluated sequentially along the horizon to obtain $\{\boldsymbol{\Sigma}_k\}_{k=0}^N$, which is then used to

compute marginal standard deviations for constraint tightening via Eq.(13) and Eq.(14). The recursion corresponds to a local (first-order) moment propagation along the MPC horizon. It provides an efficient surrogate for uncertainty evolution that avoids sampling-based stochastic MPC, while still allowing the learned residual variance to influence constraint tightening in a risk-aware manner.

Let $\mathbf{e}_k = \mathbf{x}_k - \mathbf{x}_k^{\text{ref}}$ denote the tracking error. Denote by $\mathbf{u}_k^*$ the first control input of the optimal sequence obtained from solving Eq.(15) at time step k (receding horizon). The corresponding local sensitivity of the MPC policy is

$$K_k = \left. \frac{\partial \mathbf{u}_k^*}{\partial \mathbf{x}_k} \right|_{\mathbf{x}_k = \bar{\mathbf{x}}_k} \quad (21)$$

which can be interpreted as a local feedback gain around the predicted trajectory. Using the same first-order approximation, the closed-loop tracking error approximately evolves as

$$\mathbf{e}_{k+1} = \Phi_k \mathbf{e}_k + \mathbf{w}_k \quad (22)$$

where $\Phi_k = \mathbf{A}_k - \mathbf{B}_k K_k$ is the closed-loop error dynamics matrix, $\mathbf{B}_k \left. \frac{\partial \mathbf{u}_k^*}{\partial \mathbf{u}} \right|_{(\bar{\mathbf{x}}_k, \mathbf{u}_k)}$, and $\mathbf{w}_k \sim \mathcal{N}(0, \Sigma_{\Delta, k})$ captures the stochastic impact of the residual uncertainty in a tractable manner.

3.5. Lyapunov-based mean-square boundedness under residual uncertainty

We analyze the local stochastic behavior of the UL-MPC closed-loop system around the nominal tracking trajectory using a linearized error dynamics representation. The goal is to establish a mean-square disturbance result for the tracking error under bounded residual uncertainty that aggregates dynamics mismatch and external wind disturbance. Importantly, the adaptive constraint tightening in UL-MPC affects the feasible set and safety properties of the optimization, but it does not change the form of the local closed-loop error dynamics considered in this subsection.

The analysis below characterizes the stochastic tracking error behavior induced by the receding-horizon policy whenever the MPC optimization remains feasible.

The following result is derived under a standard local stabilizability assumption commonly adopted in MPC analysis: in a neighborhood of the reference trajectory, the receding-horizon control law admits an equivalent locally stabilizing feedback representation, and the closed-loop error dynamics can be approximated by Eq.(22).

Theorem 1 (Mean-square boundedness of linearized error dynamics). Consider the linearized closed-loop tracking error dynamics Eq.(22):

$$\mathbf{e}_{k+1} = \Phi_k \mathbf{e}_k + \mathbf{w}_k$$

where $\Phi_k = \mathbf{A}_k - \mathbf{B}_k K_k$ is the local closed-loop operator defined in Section 3.4, and $\mathbf{w}_k$ is a zero-mean additive disturbance capturing the residual uncertainty, with conditional covariance $\mathbb{E}[\mathbf{w}_k \mathbf{w}_k^T | \mathcal{F}_k] = \Sigma_{\Delta, k}$ for some filtration

$\mathbf{F}_k$. Here, $\Sigma_{\Delta,k}$ captures the combined effect of dynamics-level modeling discrepancies and external disturbance represented by the one-step residual, as described in Section 3.2. Measurement noise is handled by the state estimation pipeline and is not explicitly modeled in $\mathbf{w}_k$.

Suppose there exist symmetric matrices $\mathbf{P} \succ 0$ and $\mathbf{Q} \succ 0$ such that

$$-\mathbf{Q} \pm \Phi_k^T \mathbf{P} \Phi_k - \mathbf{P}, \quad \forall k \in \mathbb{N} \quad (23)$$

and the residual covariance is uniformly bounded in the sense that

$$\text{Tr}(\mathbf{P} \Sigma_{\Delta,k}) \leq c < \infty, \quad \forall k \in \mathbb{N} \quad (24)$$

Then the tracking error is mean-square bounded and satisfies

$$\limsup_{k \rightarrow \infty} \mathbb{E}[\|\mathbf{e}_k\|^2] \leq \frac{c}{\lambda_{\min}(\mathbf{Q})} \quad (25)$$

PROOF. Consider the Lyapunov function candidate

$$V_k \triangleq \mathbb{E}[\mathbf{e}_k^T \mathbf{P} \mathbf{e}_k], \quad \mathbf{P} \succ 0 \quad (26)$$

From Eq.(22), taking conditional expectation with respect to $\mathbf{F}_k$ and using $\mathbb{E}[\mathbf{w}_k | \mathbf{F}_k] = 0$, we obtain

$$\begin{aligned} \mathbb{E}[V_{k+1} - V_k | \mathbf{F}_k] &= \mathbf{e}_k^T (\Phi_k^T \mathbf{P} \Phi_k - \mathbf{P}) \mathbf{e}_k + \text{Tr}(\mathbf{P} \mathbb{E}[\mathbf{w}_k \mathbf{w}_k^T | \mathbf{F}_k]) \\ &= \mathbf{e}_k^T (\Phi_k^T \mathbf{P} \Phi_k - \mathbf{P}) \mathbf{e}_k + \text{Tr}(\mathbf{P} \Sigma_{\Delta,k}) \end{aligned} \quad (27)$$

Applying Eq.(23) and Eq.(24) to Eq.(27) yields

$$\mathbb{E}[V_{k+1} - V_k | \mathbf{F}_k] \leq -\mathbf{e}_k^T \mathbf{Q} \mathbf{e}_k + c \leq -\lambda_{\min}(\mathbf{Q}) \|\mathbf{e}_k\|^2 + c \quad (28)$$

Taking total expectation and summing recursively gives

$$V_{k+1} \leq V_k - \lambda_{\min}(\mathbf{Q}) \mathbb{E}[\|\mathbf{e}_k\|^2] + c$$

which implies, by standard stochastic Lyapunov drift arguments, that

$$\limsup_{k \rightarrow \infty} \mathbb{E}[\|\mathbf{e}_k\|^2] \leq \frac{c}{\lambda_{\min}(\mathbf{Q})}$$

Remark 1. Theorem 1 characterizes the local mean-square boundedness of the UL-MPC closed-loop system under bounded residual uncertainty. The bound in Eq.(25) explicitly relates the residual covariance $\Sigma_{\Delta,k}$ (predicted by the probabilistic residual model) to the steady-state tracking error magnitude, illustrating how uncertainty quantification contributes to robust performance. Note that the result is derived for the linearized dynamics around the reference trajectory and relies on the existence of a locally stabilizing MPC feedback representation. Moreover, since the adaptive tightening acts on constraint handling rather than directly altering the local error dynamics, the bound describes the closed-loop tracking performance under bounded residual covariance whenever the tightened MPC problem remains feasible.

4. Experiments and results

This section presents a series of simulation experiments to evaluate the effectiveness of the proposed learning-based MPC method under model uncertainty and external disturbances. All experiments are conducted in the Gazebo simulation environment, using a high-fidelity quadrotor model with realistic physics. We compare the performance of our method with several baseline controllers, and assess both tracking accuracy and robustness under wind disturbance.

4.1. Simulation setup

The simulation environment is shown in Fig. 2. A high-fidelity quadrotor model is implemented in Gazebo. Reference trajectories are generated by fitting randomly sampled waypoints, yielding both smooth paths and aggressive maneuvers. The aggressive trajectories contain sharp turns and altitude transitions that induce rapid changes in heading and velocity, thereby stressing the controller under fast-varying effects and increased dynamics mismatch. The average reference speed is $v_{\text{mean}}^{\text{ref}}$. Each experiment is repeated 10 times with different disturbance realizations to report statistical averages.

To evaluate robustness, we apply a stochastic wind disturbance in the world frame, modeled as

$$\mathbf{d}(t) = \bar{\mathbf{d}} + \boldsymbol{\eta}(t), \quad (29)$$

where $\bar{\mathbf{d}}$ is a constant mean wind along the x or y-axis with magnitude in the range 1-2 m/s. The fluctuation term is modeled as Gaussian noise, $\boldsymbol{\eta}(t) \sim \mathcal{N}(0, \sigma_w^2 I)$. Unless otherwise stated, we set $\sigma_w = 0.1$ m/s.

The closed-loop control is executed at 50 Hz. The MPC optimization is re-solved at a lower replanning rate, and the computed thrust input is applied between replanning instants in a receding-horizon manner. The MPC horizon is $T_h = 1$ s with $N = 10$ control intervals, yielding an optimal control problem (OCP) discretization step of $\Delta t = T_h / N = 0.1$ s. The optimal control problem is solved using acados³³ via CasADi with an SQP-RTI scheme for real-time execution. The probabilistic residual model is trained offline using 20,000 transition samples collected from nominal controller rollouts under randomized wind conditions; training details and hyperparameters are provided in Appendix A.

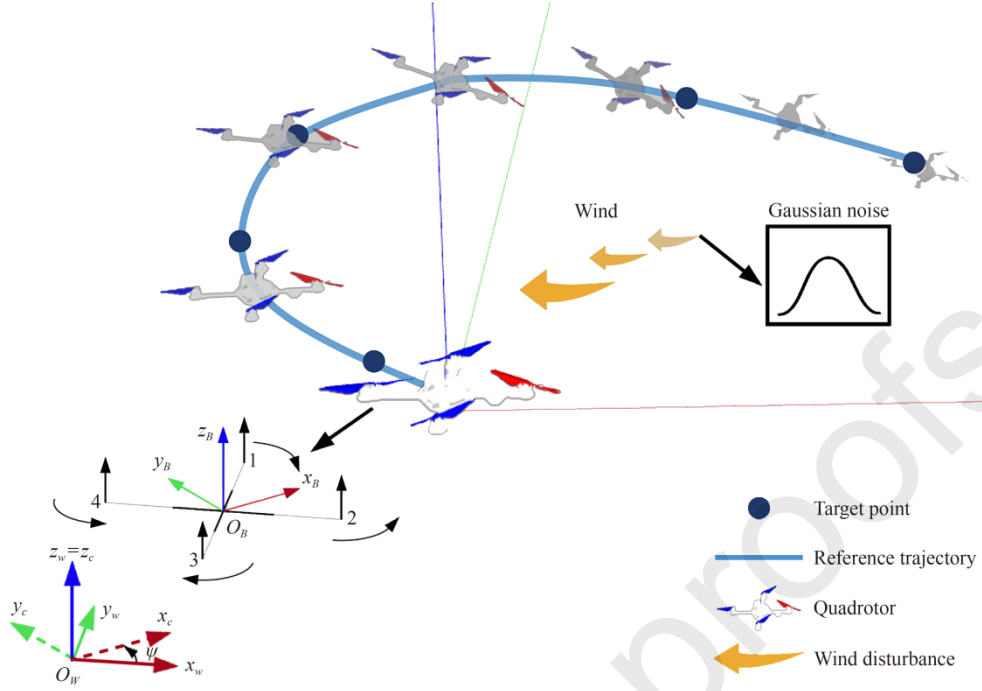


Fig. 2. Gazebo simulation scenario for evaluating trajectory tracking performance.

4.2. Baselines and evaluation metrics

To validate the effectiveness of our proposed uncertainty-aware MPC method, we compare it against several baseline approaches commonly used in learning-based control. The evaluated methods are as follows:

- **Nominal MPC:** A standard model predictive control strategy based solely on the analytical nominal model f_{nom} , without any learned correction or uncertainty compensation.
- **GP-MPC²:** A model predictive control baseline where the residual dynamics are learned using a Gaussian Process (GP), and we follow the referenced work by using only the GP posterior mean for residual compensation. The GP predictive variance is not used for constraint tightening in this baseline.
- **NN-MPC⁸:** A learning-based MPC method in which a neural network model is trained to predict the residual dynamics.
- **UL-MPC (Ours):** Our complete method, which learns the residual dynamics via a probabilistic neural network. The network outputs both the mean and variance of the residual, enabling a risk-sensitive MPC through uncertainty-aware constraint tightening.

We assess the performance of all controllers using the following quantitative metrics:

- **Tracking RMSE:** The Root Mean Square Error (RMSE) of the position tracking error between the executed position trajectory and the reference position trajectory, defined as

$$\text{RMSE} = \sqrt{\frac{1}{T} \sum_{k=1}^T \|p_k - p_k^{\text{ref}}\|_2^2} \quad (30)$$

where $p_k = [x_k, y_k, z_k]^T$ and $p_k^{\text{ref}} = [x_k^{\text{ref}}, y_k^{\text{ref}}, z_k^{\text{ref}}]^T$ denote the executed and reference position vectors in the inertial frame, respectively. The norm $\|\cdot\|_2$ represents the three-dimensional Euclidean distance between the two positions.

- **Success Rate (SR):** Fraction of trials that complete the full reference trajectory without triggering a failure event. A trial is declared successful if the quadrotor reaches the terminal reference point within the prescribed duration. A trial is declared a failure if the closed-loop simulation terminates due to a crash event, namely ground impact, corresponding to a loss-of-control episode. SR is computed over repeated runs under the same disturbance setting for each controller.
- **Constraint Violation Rate (ConVio):** Fraction of time steps violating a tubular safety-boundary around the reference trajectory. We define

$$\mathbf{X}_{\text{safe}} @ \{ \mathbf{x}_k : \| \mathbf{p}_k - \mathbf{p}_k^{\text{ref}} \|_2 \leq r_{\text{safe}} \}, \quad r_{\text{safe}} = 0.5 \text{ m} \quad (31)$$

and compute

$$\text{ConVio} = \frac{1}{T} \sum_{k=1}^T \mathbb{I}[\mathbf{x}_k \ni \mathbf{X}_{\text{safe}}] \quad (32)$$

where T is the trajectory length and $\mathbb{I}[\cdot]$ is the indicator function. The boundary radius r_{safe} is chosen to represent a practical tolerance band for tracking under wind disturbances and is kept fixed across all methods.

- **Average computation time:** The mean computation time per control step required to solve the MPC optimization problem. This metric reflects the real-time feasibility.

For completeness, we additionally report the Integrated Absolute Error (IAE) in Appendix A, which provides a complementary aggregate tracking-error measure and leads to consistent trends with RMSE.

4.3. Trajectory tracking performance

This section evaluates trajectory tracking under different disturbance conditions, reference speeds, and trajectory complexities. We focus on robustness to wind disturbance, tracking accuracy, and real-time feasibility.

4.3.1. Robustness under disturbance.

Fig. 3 presents a qualitative comparison of position tracking across multiple flight scenarios, including both low and high average speed references, with and without wind disturbances. The figure is organized with four rows, corresponding respectively to Nominal MPC, GP-MPC, NN-MPC, and the proposed UL-MPC. The four columns represent distinct operational conditions: Columns 1 and 3 (blue backgrounds) correspond to the no-wind case, while Columns 2 and 4 (orange backgrounds) correspond to a stochastic wind disturbance consisting of a mean 2 m/s wind along the x-axis with added Gaussian noise. Columns 1-2 use $v_{\text{mean}}^{\text{ref}} = 5 \text{ m/s}$ and Columns 3-4 use $v_{\text{mean}}^{\text{ref}} = 7 \text{ m/s}$. In each subplot, blue dashed lines denote the reference trajectories, colored solid lines indicate the executed trajectories, and pink segments highlight safety-boundary violations. Each subplot visualizes one representative run for trajectory illustration, whereas the annotated tracking error RMSE and ConVio values are averaged over 10 independent runs under the same condition (consistent with the quantitative tables).

Table 1 Performance comparison of different controllers under wind disturbances.

Condition	Method	$v_{\text{mean}}^{\text{ref}} = 5 \text{ m/s}$			$v_{\text{mean}}^{\text{ref}} = 7 \text{ m/s}$		
		Tracking error RMSE (mm)	$v_{\text{max}}^{\text{exe}}$ (m/s)	ConVio (%)	Tracking error RMSE (mm)	$v_{\text{max}}^{\text{exe}}$ (m/s)	ConVio (%)
No wind	Nominal MPC	377	8.54	24.3	769	10.39	58.8

	GP-MPC	162	8.38	0	576	10.57	33.4
	NN-MPC	149	8.48	0	428	9.97	15.3
	UL-MPC (Ours)	60	8.52	0	326	10.07	4.7
With wind	Nominal MPC	748	7.74	50.9	1342	10.59	72.9
	GP-MPC	703	7.83	53.2	1094	9.68	70.6
	NN-MPC	511	8.32	36.4	896	9.68	53.3
	UL-MPC (Ours)	395	8.44	25.1	611	10.94	29.6

Notes: Tracking error RMSE : root mean squared error of the position tracking error; $v_{\max}^{\text{exe}}$: maximum executed velocity;
ConVio: constraint violation rate.

Visually, Fig. 3 shows that the proposed UL-MPC achieves closer trajectory tracking compared with Nominal MPC, GP-MPC, and NN-MPC, particularly under wind disturbances. To quantify these observations, Table 1 reports the tracking error RMSE and constraint violation rate (ConVio) for references with $v_{\text{mean}}^{\text{ref}} \in \{5, 7\}$ m/s, together with the maximum executed velocity $v_{\max}^{\text{exe}}$ as an auxiliary indicator of how aggressively the controller follows the reference. Here, $v_{\text{mean}}^{\text{ref}}$ characterizes the difficulty of the tracking task, while $v_{\max}^{\text{exe}}$ denotes the maximum velocity actually sustained during flight, reflecting the ability of exploiting the agility of the reference trajectory. In the no-wind setting, UL-MPC achieves the lowest RMSE at both reference speeds and incurs nearly zero safety-boundary violations. At $v_{\text{mean}}^{\text{ref}} = 7$ m/s, UL-MPC reduces the RMSE by 57.6% relative to Nominal MPC and by 23.8% relative to NN-MPC, while maintaining a comparable $v_{\max}^{\text{exe}}$. Under wind disturbances, all baselines exhibit pronounced degradation in both RMSE and ConVio. UL-MPC remains substantially more robust. At $v_{\text{mean}}^{\text{ref}} = 7$ m/s, it reduces the RMSE by 31.8% relative to NN-MPC and lowers ConVio from 53.3% to 29.6%, indicating improved constraint satisfaction without sacrificing tracking performance under wind disturbances.

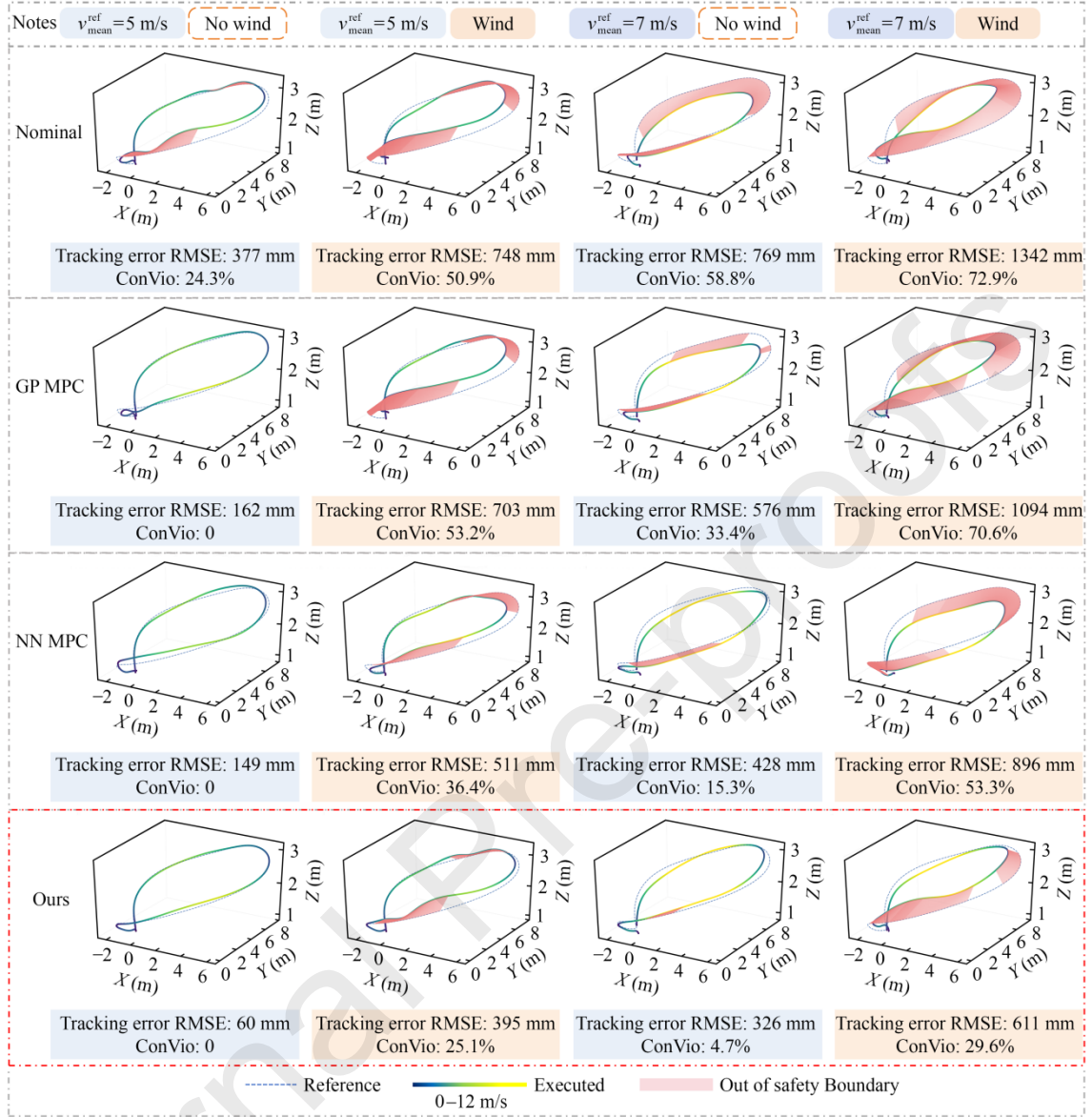


Fig. 3 Quantitative comparison of trajectory tracking performance under varying operational conditions.

In addition to the stochastic wind in Eq.(29), we include an abrupt step-gust as a distribution-shift test. Unlike the stationary stochastic wind used for training and evaluation, the step change introduces a nonstationary disturbance transient that is not represented by the steady wind statistics, and therefore probes robustness under rapid, out-of-distribution disturbance changes. Specifically, the mean wind switches from 0 to $[3, 0, 0]^T \text{ m/s}$ at $t_g = 2 \text{ s}$ while keeping the same fluctuation model as in Section 4.1. Table 2 reports tracking error RMSE, ConVio, and SR (mean over 10 runs) under this step-gust setting.

Table 2 Performance under step-gust disturbance with $t_g = 2 \text{ s}$ under $v_{\text{mean}}^{\text{ref}} = 5 \text{ m/s}$ (mean over 10 runs).

Method	Tracking error RMSE (mm)	ConVio (%)	SR (%)
Nominal MPC	1280	78.0	80
GP-MPC	1150	74.0	90
NN-MPC	930	56.0	90

Overall, these results validate the robustness and precision of the proposed UL-MPC in high-speed, disturbance-prone environments, highlighting its superiority in both tracking fidelity and dynamic performance.

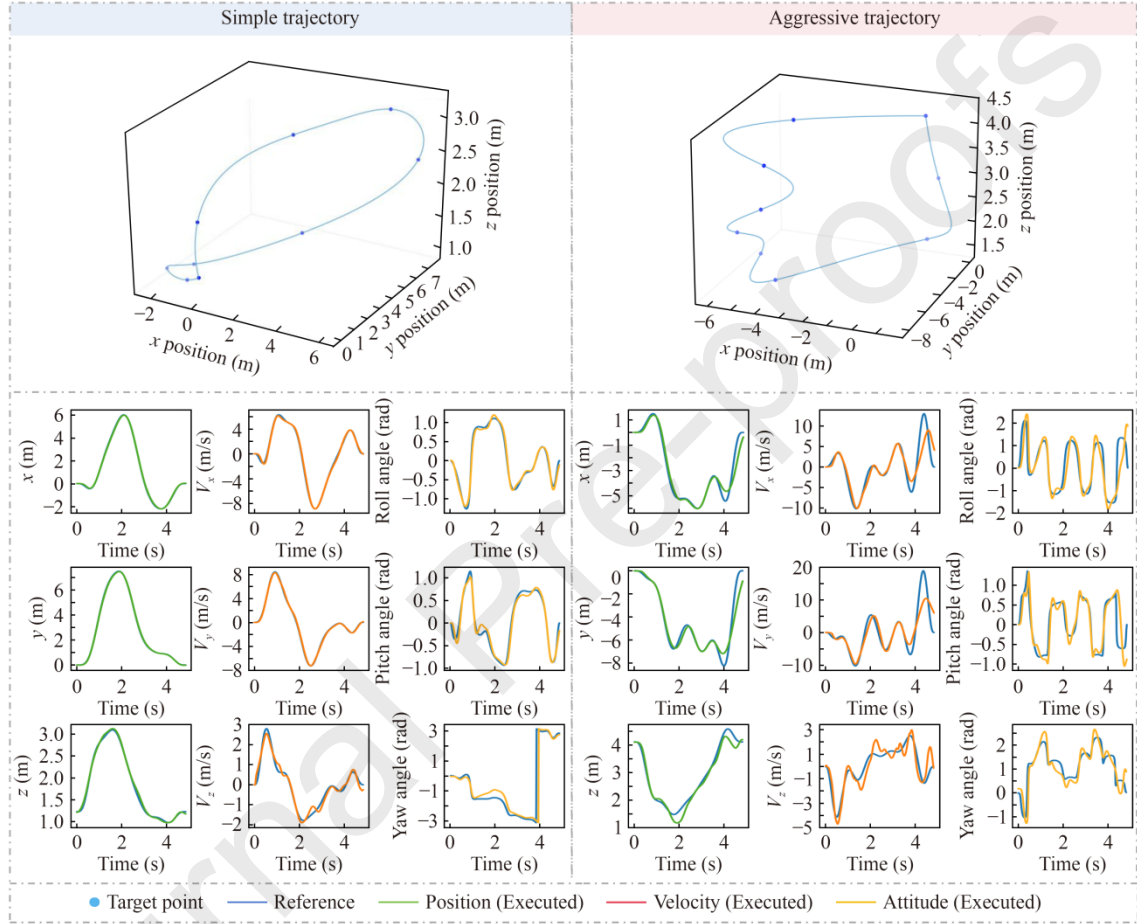


Fig. 4 Comparison of tracking performance under trajectories of varying complexity. (a) shows the 3D reference trajectory. (b) reports the position, velocity, and attitude tracking performance in x/y/z axes. Left: simple trajectory; Right: aggressive trajectory.

4.3.2. Generalization across trajectory complexity.

To further examine the consistency of the proposed controller under tasks of varying difficulty, we consider two representative test cases: (a) a simple trajectory following a low-speed circular path, and (b) an aggressive trajectory characterized by sharp turns and rapid accelerations. Both trajectories are generated through waypoint-based polynomial interpolation and are illustrated in Fig. 4. The corresponding quantitative results are summarized in Table 3. Simple refers to a low-speed circular trajectory, and Aggressive refers to a trajectory with sharp turns and rapid accelerations.

As trajectory complexity increases, all baselines show degraded performance (higher RMSE, lower SR, and higher ConVio). In contrast, UL-MPC consistently achieves the best tracking accuracy and the lowest violation rate, with SR reaching 100% in both cases, indicating improved robustness to challenging maneuvering conditions.

Table 3 Tracking performance across trajectories of increasing difficulty.

Method	Simple			Aggressive		
	Tracking error RMSE (mm)	SR (%)	ConVio (%)	Tracking error RMSE (mm)	SR (%)	ConVio (%)
Nominal MPC	779	90	50.9	1145	80	88.3
GP-MPC	633	90	53.2	943	90	80.5
NN-MPC	521	100	36.4	782	90	58.8
UL-MPC (Ours)	365	100	25.1	673	100	31.6

As the trajectory complexity increases, all baseline controllers exhibit degraded performance, manifested in larger RMSE values, reduced success rates, and more frequent violations of safety constraints. In contrast, the proposed UL-MPC consistently sustains robust tracking performance across both cases. Specifically, it achieves the lowest RMSE, maintains a 100% success rate, and incurs the fewest constraint violations, regardless of trajectory difficulty. These results highlight the strong generalization capability and disturbance resilience of the proposed approach when applied to flight tasks of varying complexity.

4.3.3. Real-time feasibility.

To evaluate the practical feasibility of deploying the proposed controller in real-time, we measure the average control frequency and computation time across all compared methods. All experiments are conducted on a CPU-based platform (Intel i7-12700H), with optimization formulated in the CasADi framework and solved using embedded numerical solvers. The reported time in Table 4 includes the complete pipeline, i.e., residual-model evaluation, covariance propagation, and the acados SQP-RTI solve. The proposed UL-MPC achieves an average computation time of 21.8 ms, which is comparable to deterministic NN-MPC and substantially faster than GP-MPC, whose data-dependent complexity leads to higher latency. In contrast, nominal MPC achieves the lowest computation time of 12.3 ms, but its lack of robustness under model mismatch limits its practical reliability. Overall, UL-MPC enables real-time implementation at 50 Hz while simultaneously ensuring robustness against disturbances.

Table 4 Average computation time.

Method	Computation time (ms)
Nominal MPC	12.3
GP-MPC	28.6
NN-MPC	21.5
UL-MPC (Ours)	21.8

4.3.4. Deployment plan and sim-to-real considerations.

Although the evaluation in this work is conducted in high-fidelity simulation, the proposed UL-MPC is designed to be directly deployable on real quadrotor platforms. In particular, the online computation comprises three deterministic components: (a) residual-model forward inference, (b) first-order covariance propagation along the horizon, and (c) solution of the underlying OCP using an RTI scheme. The timing results in Table 4 already report the end-to-end latency of this complete pipeline, indicating feasibility for a 50 Hz control loop on a standard CPU.

For onboard deployment, the control stack can follow a standard split architecture: a state estimator provides $\mathbf{x}_k$ to the UL-MPC module running on a companion computer, while the resulting thrust commands are tracked by a low-level rate controller on the flight controller. The residual model is trained offline and kept fixed during flight in the current implementation, which simplifies deployment and helps ensure predictable compute footprints.

Regarding sim-to-real generalization, the residual-learning formulation is agnostic to the data source: transition tuples $(\mathbf{x}_k, \mathbf{u}_k, \mathbf{x}_{k+1})$ collected from real flights can be incorporated using the same residual definition and NLL objective. In practice, a small amount of real-flight data can be used to fine-tune the residual model (or calibrate its uncertainty) to reduce the simulation gap, while preserving the MPC structure and constraints. A systematic hardware validation, including incremental flight tests from hover to aggressive maneuvers under controlled wind conditions, is an important direction for future work.

Across all evaluated scenarios, the proposed UL-MPC consistently outperforms baseline methods in terms of tracking accuracy, success rate, and disturbance tolerance. These results demonstrate that explicitly incorporating learned model uncertainty into the control pipeline significantly enhances robustness, thereby enabling reliable and agile quadrotor flight even under challenging real-world conditions.

5. Conclusions

This paper presented an uncertainty-aware learning-based model predictive control (UL-MPC) framework for robust trajectory tracking of quadrotors under agile flight conditions with unmodeled dynamics and wind disturbances. The proposed approach integrates residual dynamics compensation with uncertainty estimation via a probabilistic dual-head neural network. By augmenting the nominal system dynamics with data-driven corrections and embedding uncertainty into a chance-constrained MPC formulation, the controller enables risk-sensitive constraint handling under uncertainty.

Extensive simulations on aggressive and disturbance-rich flight tasks demonstrate that UL-MPC improves tracking accuracy and constraint satisfaction compared to both nominal MPC and learning-based baselines. The results confirm improved robustness across varying trajectory complexities and disturbance conditions, with computation times compatible with the target control rate.

Future research will focus on extending this framework to onboard deployment on real quadrotor platforms and establishing formal stability guarantees for uncertainty-aware learning-based control. Another promising direction is to integrate adaptive data collection and safe online updates of the residual model, enabling sustained performance under distribution shifts and complex environmental disturbances.

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References

1. Bauersfeld L, Kaufmann E, Foehn P, et al. NeuroBEM: Hybrid aerodynamic quadrotor model. 2021 Robotics: Science and Systems (RSS). 2021.
2. Torrente G, Kaufmann E, Föhn P, et al. Data-driven MPC for quadrotors. IEEE Robot Autom Lett 2021;6(2):3769–76.

3. Mehndiratta M, Kayacan E. Gaussian process-based learning control of aerial robots for precise visualization of geological outcrops. 2020 European control conference (ECC). Piscataway: IEEE Press; 2020.
4. Saviolo A, Li GR, Loianno G. Physics-inspired temporal learning of quadrotor dynamics for accurate model predictive trajectory tracking. *IEEE Robot Autom Lett* 2022;7(4):10256–63.
5. Ventura Diaz P, Yoon S. High-fidelity computational aerodynamics of multi-rotor unmanned aerial vehicles. 2018 AIAA aerospace sciences meeting. Reston: AIAA; 2018.
6. Hewing L, Wabersich KP, Menner M, et al. Learning-based model predictive control: Toward safe learning in control. *Annu Rev Control Robot Auton Syst* 2020;3:269–96.
7. Song YL, Scaramuzza D. Policy search for model predictive control with application to agile drone flight. *IEEE Trans Robot* 2022;38(4):2114–30.
8. Salzmann T, Kaufmann E, Arrizabalaga J, et al. Real-time neural MPC: Deep learning model predictive control for quadrotors and agile robotic platforms. *IEEE Robot Autom Lett* 2023;8(4):2397–404.
9. Saveriano M, Yin Y, Falco P, Lee D. Data-efficient control policy search using residual dynamics learning. 2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). Piscataway: IEEE Press; 2017.
10. Spielberg NA, Brown M, Gerdes JC. Neural network model predictive motion control applied to automated driving with unknown friction. *IEEE Trans Control Syst Technol* 2022;30(5):1934–45.
11. Wang YR, Boyle D. Constrained reinforcement learning using distributional representation for trustworthy quadrotor UAV tracking control. *IEEE Trans Autom Sci Eng* 2025;22:5877–94.
12. Hamza A, Mohamed AH, El-Badawy A. Robust H-infinity control for a quadrotor UAV. AIAA SCITECH 2022 forum. Reston: AIAA; 2022.
13. Abdelhay S, Zakriti A. Modeling of a quadcopter trajectory tracking system using PID controller. *Procedia Manuf* 2019;32:564–71.
14. Benevides JRS, Paiva MAD, Simplício PVG, et al. Disturbance observer-based robust control of a quadrotor subject to parametric uncertainties and wind disturbance. *IEEE Access* 2022;10:7554–65.
15. Kharisov E, Kim KK, Hovakimyan N, et al. Limiting behavior of L1 adaptive controllers. AIAA guidance, navigation, and control conference. Reston: AIAA; 2011.
16. Kabzan J, Hewing L, Liniger A, et al. Learning-based model predictive control for autonomous racing. *IEEE Robot Autom Lett* 2019;4(4):3363–70.
17. Hewing L, Kabzan J, Zeilinger MN. Cautious model predictive control using Gaussian process regression. *IEEE Trans Control Syst Technol* 2020;28(6):2736–43.
18. Chua K, Calandra R, McAllister R, et al. Deep reinforcement learning in a handful of trials

using probabilistic dynamics models. arXiv:1805.12114;2018.

19. Gal Y, Ghahramani Z. Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. International conference on machine learning. 2015.

20. Kahn G, Villaflor A, Pong V, et al. Uncertainty-aware reinforcement learning for collision avoidance. arXiv:1702.01182; 2017.

21. Kulathunga G, Hanheide M, Klimchik A. Sparse Gaussian process regression for residual dynamics learning in multi-rotor aerial vehicles control. 2024 IEEE 20th international conference on automation science and engineering (CASE). Piscataway: IEEE Press; 2024.

22. Qiu D, Zhao Y, Baker CL. Latent belief space motion planning under cost, dynamics, and intent uncertainty. 2020 Robotics: Science and Systems (RSS) 2020.

23. Sanyal S, Roy K. RAMP-net: A robust adaptive MPC for quadrotors via physics-informed neural network. 2023 IEEE international conference on robotics and automation (ICRA). Piscataway: IEEE Press; 2023.

24. Nie WM, Li HF, Zhang R. Model-free adaptive optimal design for trajectory tracking control of rocket-powered vehicle. Chin J Aeronaut 2020;33(6):1703–16.

25. Kulathunga G, Hamed H, Klimchik A. Residual dynamics learning for trajectory tracking for multi-rotor aerial vehicles. Sci Rep 2024;14:1858.

26. Saviolo A, Frey J, Rathod A, et al. Active learning of discrete-time dynamics for uncertainty-aware model predictive control. IEEE Trans Robot 2024;40:1273–91.

27. Shi GY, Shi XC, O’Connell M, et al. Neural lander: Stable drone landing control using learned dynamics. 2019 international conference on robotics and automation (ICRA). Piscataway: IEEE Press; 2019.

28. Li JJ, Han L, Yu HY, et al. Nonlinear MPC for quadrotors in close-proximity flight with neural network downwash prediction. 2023 62nd IEEE conference on decision and control (CDC). Piscataway: IEEE Press; 2024.

29. Depeweg S, Hernández-Lobato JM, Doshi-Velez F, et al. Decomposition of uncertainty in Bayesian deep learning for efficient and risk-sensitive learning. International conference on machine learning., 2017

30. Zheng L, Yang R, Pan JS, et al. Learning-based safety-stability-driven control for safety-critical systems under model uncertainties. 2020 international conference on wireless communications and signal processing (WCSP). Piscataway: IEEE Press; 2020.

31. Abraham I, Murphey TD. Active learning of dynamics for data-driven control using Koopman operators. IEEE Trans Robot 2019;35(5):1071–83.

32. Andersson JAE, Gillis J, Horn G, et al. CasADi: A software framework for nonlinear optimization and optimal control. Math Program Comput 2019;11(1):1–36.

33. Verschueren R, Frison G, Kouzoupis D, et al. Acados—A modular open-source framework

for fast embedded optimal control. Math Program Comput 2022;14(1):147–83.

Appendix A. Training details and additional metric

A1. Training configuration

The probabilistic residual network is trained offline using 20000 one-step transition samples collected from nominal MPC rollouts under randomized trajectories and randomized wind disturbance conditions. Table A.1 summarizes the network architecture and the training hyperparameters used in this work.

Table A1 Training hyperparameters for the probabilistic residual network.

Item	Setting
Network backbone	MLP with 3 hidden layers (64 units each), ReLU activations
Dual-head outputs	mean head: $\mu_{\Delta, \theta}$; variance head: s_{θ}
Variance mapping	$\ln \sigma^2 = s$, $\sigma^2 = \exp(s) + \sigma_{\min}^2$
$\sigma_{\min}^2$	10^{-6}
Optimizer	Adam
Learning rate	1×10^{-4}
Batch size	128
Max epochs	200
Weight decay (l_2)	1×10^{-4}
Early stopping	patience = 20 epochs (based on validation NLL)

A.2. Integrated Absolute Error (IAE)

In addition to tracking error RMSE, we report the Integrated Absolute position Error (IAE) in position as a complementary tracking metric:

$$\text{IAE} @ \sum_{k=0}^K \| \mathbf{p}_k - \mathbf{p}_k^{\text{ref}} \|_2 T_s \quad (\text{A1})$$

where $\| \mathbf{p}_k - \mathbf{p}_k^{\text{ref}} \|_2$ is the position error magnitude (in m), T_s is the closed-loop sampling period (in s), and K is the number of samples in one trial. The summation is a discrete-time approximation of the time integral of the position error over the trial, and IAE is reported in m · s

Table A2 The position tracking IAE under the same conditions as Table 1 (mean over 10 runs).

Condition	Method	$v_{\text{mean}}^{\text{ref}} = 5 \text{ m/s}$	$v_{\text{mean}}^{\text{ref}} = 7 \text{ m/s}$
No wind	Nominal MPC	1.07	2.25
	GP-MPC	0.46	1.42
	NN-MPC	0.33	1.26
	UL-MPC (Ours)	0.19	0.98

	Nominal MPC	2.17	4.03
With wind	GP-MPC	2.05	3.32
	NN-MPC	1.46	2.84
	UL-MPC (Ours)	1.07	2.01

Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as

potential competing interests: